CHAPTER

11

Isolation of the Operating Field

Any operative procedure necessitates the need for adequate control over the operating field. It is imperative that there should be proper moisture control, good accessibility and visibility as well as adequate space for instrumentation around the working area. Such an environment is necessary for easy manipulation of the restorative materials. Isolating the working area includes isolation from moisture like saliva, blood and gingival crevicular fluid and isolation from the soft tissues like lips, cheeks, gingiva and tongue. A number of methods can be employed either singly or in combination to obtain isolated environment for the operator. Isolation is studied under two heads:

I. Isolation from moisture
II. Isolation from the soft tissues

I. ISOLATION FROM MOISTURE

Various aids available for this purpose are:

A. Direct methods
   a. Rubber dam
   b. Cotton rolls and cotton roll holder
   c. Gauze pieces
   d. Absorbent wafers
   e. Suction devices (evacuation system)
      i. High volume evacuator
      ii. Low volume evacuator
   f. Gingival retraction cord
B. Indirect methods
   a. Comfortable position of the patient and relaxed surroundings
   b. Local anesthesia
   c. Drugs
      i. Antisialogogues
      ii. Antianxiety drugs
      iii. Muscle relaxants

A. DIRECT METHODS

a. Rubber Dam

Dr SC Barnum (1864) introduced rubber dam, one of the best methods for providing isolation from saliva and soft tissues. Initially, the placement of rubber dam was cumbersome; however, with the improved varieties of rubber dam, the process is simple and effective.

Purpose

- Isolation of the operating site from moisture, i.e. saliva, sulcular fluid and blood.
- Retraction of the soft tissue, i.e. cheeks, lips, tongue and minimally, gingiva.
- Increases accessibility and visibility to the working area.
- Improves efficiency of the operator (intermittent expectoration and rinsing by the patient is avoided).
- Improves properties of the dental materials and hence the final outcome of the restoration.
- Protection of the patient and the operator. The patient is protected against accidental aspiration of instruments, debris, medicaments or irrigating solutions. Protects against soft tissue injury when using rotary instruments. Additionally, both the patient and the operator are protected against any cross-infection.
- Reduces chair time for the restoration.

Use of rubber dam should be avoided in:

- Asthmatics and mouth breathers; they may not tolerate the dam.
- Partially erupted and mal-positioned teeth; may not receive a retainer easily.
- Psychological reasons (personal disliking).

Armamentarium

A rubber dam kit (Fig. 11.1A) has the following items:

a. ***Rubber dam sheets:*** Available in the form of rolls from which square sheets can be cut; individual sheets are also available (Fig. 11.1B, C). The characteristics of rubber dam sheath are:
   - *Size:* 5"× 5" or 6" × 6" square
   - *Thickness:*
     - Thin—0.0063"
     - Medium—0.008"





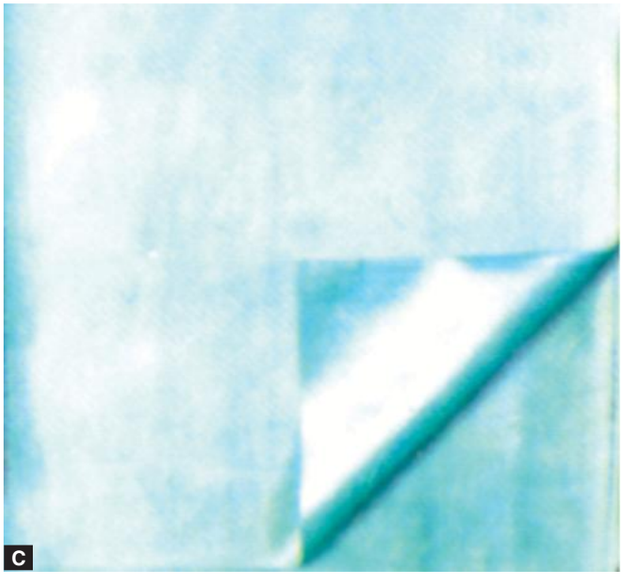


Fig. 11.1: **(A)** Rubber dam kit; **(B and C)** Rubber dam sheaths

- Heavy—0.010" (provides better retraction of soft tissue and is more resistant to tearing)
- Extra heavy—0.012"
- Special heavy—0.014"

- *Color:* Available in several colors; however, green and blue colors are preferred, because they provide good contrast with the surroundings. Rubber dam sheet has a shiny side and a dull side. The dull side should face the operator, so as to reduce any light reflected from it.

b. ***Rubber dam clamps (retainers):*** These are used to secure the dam to the teeth, which need to be isolated. Several clamps are available in various sizes and shapes. The larger clamps are used for adult patients and the smaller ones for children (Fig. 11.2). Clamps can retract the gingival tissue (though not adequate); especially useful when preparing and restoring class V cavities. A retainer has 2 jaws connected by a bow, 2 prongs on each jaw; means that there are 4 prongs in a clamp and each prong rests on the mesial/distal line angle of the tooth to be clamped. A prong should not extend beyond the angle of the tooth, otherwise it would interfere with the placement of a wedge or matrix band and may cause gingival trauma. Certain retainers have prongs that are inverted, i.e. directed gingivally. These are more convenient to use on partially erupted teeth or when additional soft tissue needs to be retracted.

The retainers are of two types:

i. *Winged retainers:* These retainers have wing-like projections on the outer aspect of their jaws; provide extra retraction of the rubber dam from the field of operation. The wings are passed through the punched hole in the dam; then the dam and retainer are placed onto the concerned tooth. After placement, the dam is slipped carefully over the wings onto the tooth.

ii. *Wingless retainers:* These retainers have no wings on their jaws; smooth on their outer aspect. The retainer is first placed on the tooth and the dam is stretched over the clamp onto the tooth.



Fig. 11.2: Rubber dam clamps

Modified designs: The clamps with modified design are:

i. Tiger clamp
- Clamp with serrated jaws
- Used for partially erupted and structurally compromised teeth

ii. Silker-Glickman clamp (S-G clamp)
- Clamp with extended wings
- Allow rubber dam placement around teeth with minimal tooth structure

iii. Haller clamp
- Holding of the tongue and cheek
- Improved relative dryness, when rubber dam is not required.

iv. Cushee clamp
- Increases patient comfort; eliminate contact of steel clamp with gingiva and tooth enamel.
- Enhances rubber dam seal; limits leakage from the dam.
- Reduces clamp slippage.

c. ***Rubber dam retainer forceps:*** It is a forceps that holds the retainer and facilitates its placement and removal from the tooth (Fig. 11.3).

d. ***Rubber dam holder (rubber dam frame):*** It holds the borders of the dam and positions it. The frame is U shaped and could be an adult or a pedodontic one made of metal or plastic. The metal one is known as the Young's frame. Plastic frame is useful when a radiograph is to be taken without removing the frame (Fig. 11.4). The frame has minute projections on its outer surface where the dam is secured. An additional two hooks may be present on the sides of the frame, where the neck strap may be optionally attached. The frame is preferably placed beneath the dam rather than above it.

Modified designs: The frames with modified design are:

i. Nygaard-Ostby frame
- Known as shark mouth
- Radiolucent nylon frame
- Polygonal in shape

ii. Articulated frame
- Foldable metal frame
- Developed to facilitate endodontic radiography

iii. Derma frame
- Pliable metal rubber dam frame
- Can be bent to take radiographs
- Patient comfort while retaining the dam in place.

iv. Safe-T-frame
- Replaces the conventional one piece frame with a two-piece frame design.
- Easy to use (more secure fit).

e. ***Rubber dam punch:*** It is a punch for making holes in the dam, characterized by a rotating metal disc, which bears five or six holes of different sizes, and a sharp pointed plunger (Fig. 11.5). When the handle of the punch is pressed, the plunger should rest in the center of the hole. The holes are of different sizes according to the size of different teeth. The smallest hole is for incisors, followed by canine; pre-molars, molars; the largest hole is for saliva ejector. Use the particular hole suggested for that particular tooth, otherwise the dam may tear during its placement.

f. ***Rubber dam template/stamp:*** The template has marked positions of the teeth. The markings are used to transfer position to the rubber dam sheet for punching the holes (Fig. 11.6A to C).

g. ***Dental floss:*** A strand of dental floss (Fig. 11.7A) should be tied around the retainer before it is carried into the oral cavity (Fig. 11.7B). This is a safety measure to prevent accidental aspiration of the clamp.



Fig. 11.3: Clamp forceps



Fig. 11.4: Rubber dam frame



Fig. 11.5: Rubber dam punch

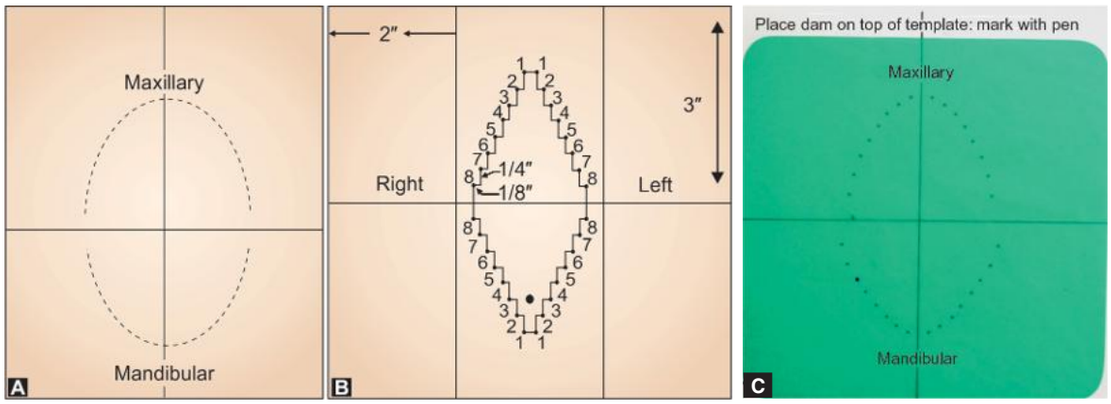


Fig. 11.6: Rubber dam template: **(A)** As provided by the manufacturer; **(B)** Self-made; **(C)** Marking over the rubber dam sheath to be placed over template customised



Fig. 11.7A: Dental floss

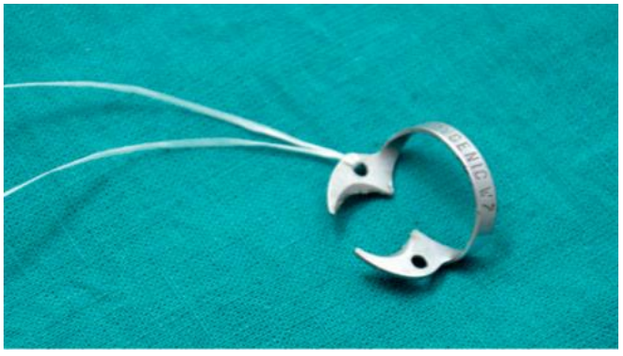


Fig. 11.7B: Dental floss tied around the retainer

The floss should be adequately long, say 12" so that the strand hangs out of the mouth for a sufficient distance. Dental floss may also be used for passing the rubber dam sheet through interproximal contact and also to serve as a retainer in place of conventional clamps.

h. *Wedjet:* This is an elastic cord, generally used to secure the dam around teeth farthest away from the clamp (Fig. 11.8). It can also be used to push the dam through the interproximal contact and also in some places as a retainer instead of a clamp.



Fig. 11.8: Stabilizing cord

i. *Lubricant:* A lubricant aids in passing the dam over the tooth. It is applied on both sides of the dam in the area of punched holes. Lubricants may be commercially available or ones like soap, vaseline, shaving cream, etc. can also be employed. Vaseline or petroleum jelly should also be applied on the patient's lips and corners of the mouth to avoid constant irritation from the rubber dam and cracking of the skin.

New Rubber Dam

a. *Instidam*

- Compact design, fits outside the patients' oral cavity.
- Built-in flexible frame, with prepunched hole, off-center by ½ inch.
- Prepunched hole helps eliminate tearing; additional holes may be punched, if necessary.
- Made with translucent natural latex; stretchable, tear resistant and provides easy visibility.
- Radiographs may be taken without removing the dam by bending the instidam.
- Produces minimal pull on clamp.
- Single use only.
- Easy manipulation and comfortable.

b. *Handidam*

- Preframed rubber dam.
- Easy to apply and saves time.

c. *OptraDam*

- Anatomical shape and integrated frame
- Placement fast and easy (Fig. 11.9).
- Flexible in all directions; comfortable for patient for long durations.
- Both arches are fully exposed and provides easier access to the treatment field.
- No metal clamps are required.
- Can be kept in place while X-rays are being taken.
- Available in two sizes—regular and small.

d. *OptiDam*

- 3-D anatomically contoured thermoplastic frame
- Allows easy placement; accommodates anatomical variations in the oral cavity
- Minimal tension on the clamp due to the design.
- Powder-free dam; reducing the chance of air-borne particles that can cause an allergic reaction.
- Available in anterior and posterior versions.
- No need of tooth marking and punching.
- Safe to autoclave at 134°C.

e. *Dry dam*

- An alternative type of rubber dam, which does not require a frame
- Fits like a face mask
- Absorbent lining provides comfort to the patient
- Non-allergic
- Useful in anterior teeth; not good for posterior teeth
- Not useful during bleaching

f. *Flexi dam* (Fig. 11.10)

- It is non-latex flexi dam available with built-in frame
- Good tear resistance and is odorless
- Plastic frame provides requisite comfort to the patient



Fig. 11.9: OptraDam



Fig. 11.10: Flexi dam

Alternative Rubber Dams

a. *Kool dam:* It is a light-cured material applied on gingiva and adjacent teeth prior to power bleaching and sand blasting providing the requisite isolation. This is also known as liquid rubber dam. It is flexible, showing good tear resistance and is easy to remove. It is user friendly rubber dam substitute (Fig. 11.11).
OpalDam, a resin-based product, is not used these days because the heat produced during curing of resin cause discomfort/pain to the patient.

b. *Fast dam:* Fast dam can be anatomically shaped providing requisite dry field for the operator. All saliva ejectors can be adjusted with the suction holes provided along the perimeter. It effectively retracts the tongue and the cheeks, maintaining a continuous dry field (Fig. 11.12).

c. *Isolite:* A new device, provides illumination, retraction, isolation and throat protection. It has soft and flexible mouthpiece, which isolates maxillary and mandibular quadrants simultaneously, providing isolation and the illumination (Fig. 11.13). It is preferred in young patients with incompletely erupted teeth.
The disadvantage of this device is being more expensive than rubber dam, not providing the color



Fig. 11.11: Kool dam



Fig. 11.12: Fast dam



Fig. 11.13: Isolite

contrast and may damage gingiva. (Isolite does not seal the gingiva from irrigants and intracanal medicaments.)

Supplementary Aids for Retention

Low fusing impression compound is occasionally used to supplement the retention of the retainer on the tooth. It is especially useful when coronal tooth structure is not adequate.

Rubber Dam Application

The important features to be considered during rubber dam application are:

- Isolate at least three teeth at a time (avoid single tooth isolation).
- For working on central incisors, lateral incisors or on mesial aspect of canines, isolate first premolar from one side to first premolar of the opposite side.
- For working on the distal aspect of canines and premolars, isolate two teeth posteriorly and two anteriorly for convenience.
- For working on molars, isolate from posterior most tooth to the premolars.
- Spacing between two holes in the dam should be adequate (approximately ¼th''). If spacing is inadequate, the rubber dam sheet may move to the mesial/distal side of the papilla, thereby exposing and injuring the gingiva. This also increases the chances of tear of the dam. If the holes are overspaced, rubber dam will bunch in between the teeth thus interfering with the operative procedure.
- When cervical retainer is to be applied, the hole for particular tooth is punched a little facial to the arch, so as to accommodate for the extension of the dam gingivally.
- When thin sheets are used, the holes punched should be of the smaller size so as to obtain proper fitting around the tooth.

The routinely used techniques for rubber dam application are:

a. ***Bow technique:*** The bow technique can be employed without an assistant. It provides a good view of the oral cavity as the clamp is being placed. A disadvantage is that the rubber must be stretched over the wings, which may lead to its tear. The largest punch should be used for making the hole to avoid tearing.

b. ***Wing technique:*** In this technique, the rubber dam is to be inserted along both wings of the clamp into the hole punched in the dam. The preliminary steps can be accomplished by the assistant and the placement is completed by the operator. The assistant inserts the selected clamp into the hole in the rubber sheet at an angle of 45° with the bow towards the distal, relative to the dental arch. Then the attachment to the rubber dam frame is carried out stretching the rubber diagonally.

The disadvantage of this technique is the reduced visibility, especially on the more distal teeth. In addition, care must be taken not to pinch or injure soft tissues, such as tongue, lips, or cheeks. As the clamp is carefully placed into undercut areas of the tooth, the patient is instructed to signal, if there is any discomfort in the gingiva. Once the rubber dam is released from the wings of the clamp, the placement is completed.

c. ***Rubber sheet first technique:*** The rubber dam clamp and forceps should be kept ready and the rubber sheet is stretched over the frame. A special rubber dam is provided with frame, which has guiding grooves for a saliva ejector at the lower right and left borders. The next step is fitting of the clamp. If it fits well, the rubber dam attached to frame is carried to the oral cavity. The hole may be spread wider with the fingers, and the rubber is pulled down around the tooth. The clamp is then placed over the isolated tooth (Fig. 11.14A, B).

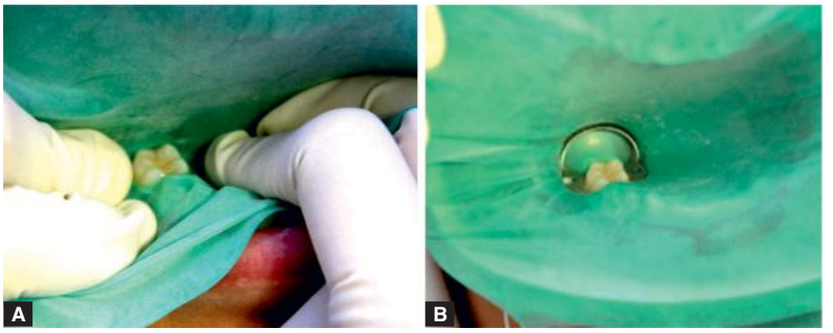


Fig. 11.14: Rubber dam sheet first technique

The advantage of this procedure is that all types of clamps can be used and the visibility is better.

d. *Clamp first technique:* The clamp is placed first on the tooth. The clamp-forceps and rubber sheet-frame should be kept ready (Fig. 11.15A–E). The order in which they are brought into the oral cavity is reversed. It is essential that the clamp be secured against the possibility of its slipping into the pharyngeal cavity.

This is accomplished by looping a long piece of dental floss over the bow of the clamp and letting it hang out of the mouth where it can be quickly



Fig. 11.15: Clamp first technique:**(A)** Punching hole in rubber dam sheet; **(B)** Application of lubricant; **(C)** Placement of rubber dam sheet over the clamp; **(D)** Adjusting rubber dam and clamp; **(E)** Completion of rubber dam placement

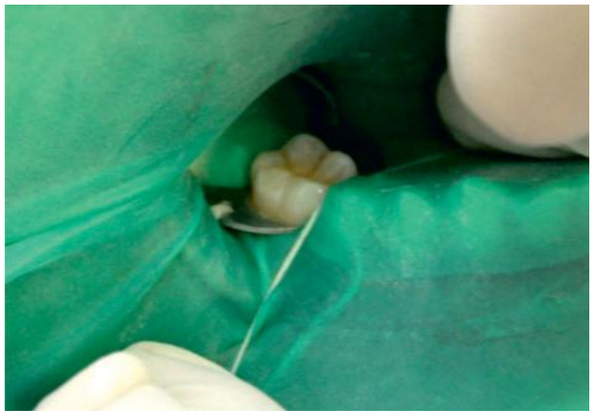


Fig. 11.16: Placing rubber dam by using floss ligatures

grasped in case of emergency. The problem generally faced during placement of the rubber dam in that the hole must be stretched wide to pass over the selected clamp. Therefore, the use of wingless clamps is recommended because the smaller width makes it less susceptible to tear. The advantage is the unobstructed view as the clamp is being applied. In a few cases, the techniques described above will have to be modified in order to achieve a well-placed, tightly fitting rubber dam. Patients with fixed orthodontic appliances pose problems. In such situations, it is possible to place a rubber dam by using dental floss ligatures (Fig. 11.16). Teeth with fixed prostheses may need endodontic treatment. If only single tooth is involved, it is better to remove the crown to obtain a better view. If the underlying tooth is severely damaged, it may be necessary to attach the dam to the adjacent teeth or to employ a clamp designed especially for deeply damaged teeth. The final step for all the techniques is to disinfect the operating field. Chlorhexidine/betadine is suitable disinfecting agent for dental use.

Procedure

The stepwise procedure of rubber dam application is as follows:

- Make the patient comfortable. Remove any debris or calculus around the teeth before placing the rubber dam. If rubber dam is to be placed around a tooth, which is restored with a proximal restoration or a crown, check for any overhanging margins or sharp edges. These are corrected before rubber dam placement to avoid tear of dam.
- Check for tightness of the proximal contacts by passing the floss from buccal or lingual. Tight proximal contacts may not allow the passage of rubber dam; consider other forms of isolation in such cases. However, minimally tight contacts that are difficult to floss can be wedged apart to allow the passage of rubber dam.
- Select a rubber dam clamp depending on the type of the tooth to be isolated. It should be stable without hurting the surrounding tissues and the restoration. A clamp forceps is used to seat the clamp onto the tooth; first on the lingual cervical region, then onto the buccal cervical region. Before trying the clamp, dental floss is tied around it. The length of the floss should be sufficiently long. Check for the stability of the clamp; slight instability may be overcome by using impression compound to stabilize the clamp.
- Take a rubber dam sheet. Punch a hole on its upper right corner or mark it with 'R' for identifying the patient's right side. The sheet is then placed on a template and the position of the holes marked on it with a pen. (With a template, only fixed positions can be obtained.) When variations are required, manual template can be utilized.
- A lubricant is applied on both sides of the punched hole to facilitate the passage of dam over the tooth. The patient's lips and the corners are also coated with a lubricant (Fig. 11.17).
- The rubber dam is now placed on the tooth. The placement can follow different patterns like:
  i. First place wingless clamp on the tooth. Stretch the lips of the appropriate hole in the rubber dam sheet and then slide it over the bow and jaws of the clamp and around the cervical of the tooth.
  ii. Place the wingless clamp and the rubber dam together around the tooth; the rubber dam is passed over the bow of the clamp. The clamp forceps engages the holes in the jaws beneath the dam and is used to place the clamp along with the dam over the tooth. The dam is stretched and slid over the jaws onto the cervix of the tooth.
  iii. Use of winged clamp: Both the clamp and the dam can be placed together; the rubber dam is passed over the bow. The clamp forceps engages the holes in the jaws and is used to place the



Fig. 11.17: Coating patient's lips with lubricant

clamp along with the dam over the tooth. The dam is stretched and slid onto the cervix of the tooth.

iv. First place the rubber dam on the tooth and then secure it in position by placing a winged or wingless clamp.

- The rubber dam should cautiously passed through the contact. First try and insert the sheet without a floss. The floss aids in slipping the edge of the dam along the tooth. Once it has passed through the contact, the contact is sufficient to hold the sheet back. Never place floss wholly on the rubber dam as it will create a double thickness of the sheet in that area.
- Similarly, the rubber dam is passed around each tooth, one by one, until the desired number of teeth have been isolated.
- The rubber dam is then bunched in one hand and with the other hand a napkin drawn over the bunched portion onto the face.
- The rubber dam is unfolded and spread neatly. Slowly and steadily, the dam is hooked to the projections on the frame, while making sure there are minimal folds in the dam. The frame can be placed either above or below the dam; preferably beneath the dam.
- The edges of the rubber dam around the holes are inverted into the gingival sulcus to obtain a proper seal around the tooth. An air-blast is used to dry the dam.
- When any operative procedure is being carried out, a low volume evacuator tip may be passed through an extra hole made near the lingual sulcus. A high volume evacuator tip can be placed above the rubber dam, if need be.
- While removing the rubber dam, pull the dam away from the tooth and cut separated interseptal rubber, if any. The retainer is removed from the tooth using a retainer forceps. The dam and the frame are then simultaneously removed from the oral cavity. The patient's lips and corners of the mouth are wiped of the lubricant with a napkin. Ask the patient to rinse and check for any left-over shreds of rubber dam. Gently massage the gingiva surrounding the tooth especially around the clamped tooth.

  Different areas isolated with rubber dam are shown in Fig. 11.18A–D.

b. Cotton Rolls and Cotton Roll Holder

Cotton rolls are moisture absorbents, and also aid facilitate retracting the soft tissues from the operating field. When used in association with profound anesthesia, cotton rolls provide acceptable dryness for procedures like examination, impression taking, cementation, sealant placement, topical fluoride application, etc. Use of a saliva ejector and cellulose

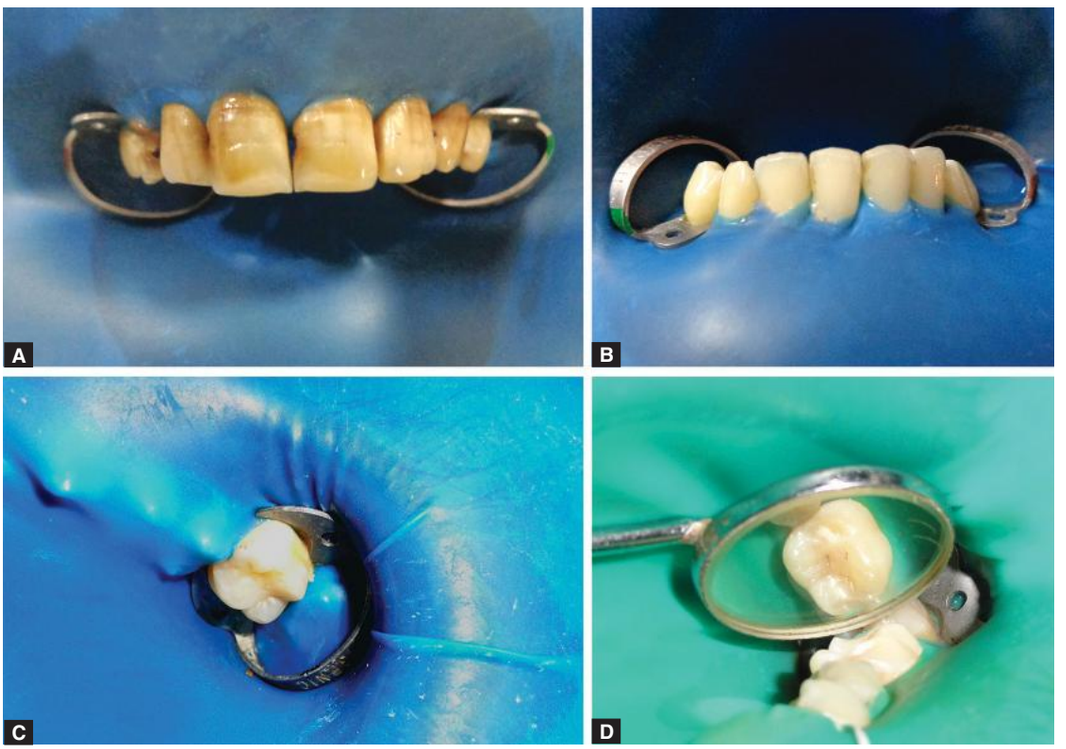


Fig. 11.18: Areas isolated with rubber dam: **(A)** Maxillary anterior teeth; **(B)** Mandibular anterior teeth; **(C)** Maxillary first molar; **(D)** Mandibular posterior teeth

wafer along with cotton rolls further control salivary flow.

The placement of cotton rolls is usually carried out by the operator's assistant. He should continually remove drenched cotton rolls and insert dry ones. Loose cotton can be rolled manually into a cotton roll; prefabricated cotton rolls are also available (Fig. 11.19). Prefabricated rolls are more compact and can absorb a greater amount of moisture.

They are available in varying diameters and lengths; usually no. 2 (small) and no. 3 (medium) sizes. The surface of the cotton roll can be smooth or woven to improve their compactness.

Cotton rolls can be placed into position and stabilized with cotton roll holders. The disadvantage with the use of cotton roll holders is that they have to be removed from the oral cavity for changing the cotton rolls. The advantage is that they provide slightly more retraction of the cheeks, lips and tongue; thus, improving accessibility and visibility of the working area.

For isolating maxillary anterior area, small-sized rolls are placed on either side of the labial frenum. For isolating mandibular anterior area, the cotton roll is placed in lingual sulcus along with one roll on either side of the mandibular labial frenum.

The maxillary posterior teeth are isolated by inserting a cotton roll in the adjacent vestibule. This will not occlude the parotid duct opening; hence, a cheek pad or cellulose wafer is additionally laid over this area of the cheek. Such placements are aided by the use of saliva ejectors to remove saliva from the lingual sulcus.

The mandibular posterior teeth are isolated by inserting one cotton roll in the buccal vestibule; usually the medium-sized roll and the larger one between the teeth and the tongue.

In the lingual sulcus area, two cotton rolls can be used, one inserted to the depth of the sulcus and the other one laid above it. Cheek pads should be applied and use of saliva ejector in the opposite lingual sulcus aids in completing the isolation.

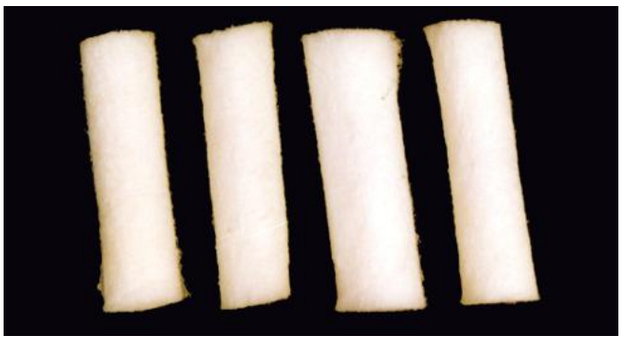


Fig. 11.19: Prefabricated cotton rolls

Avoid removing dry cotton rolls. They should be slightly moistened before removal to prevent pulling of the epithelial covering of the mucosa along with it.

c. Gauze Pieces

Gauze sponges are mainly supplied in pieces of 2" × 2" (Fig. 11.20). They perform the same function as cotton rolls and are generally used for isolating larger areas. Additionally, they may be used as throat screens when minute instruments are being used without rubber dam or when indirect restorations are being inserted, so as to avoid accidental aspiration (Fig. 11.21). The gauze sponges are better tolerated by the soft tissues, more acceptable and have less chances of adhesion to dry tissues.

d. Absorbent Pads/Wafers

Absorbent pads are generally made up of cellulose; also called cellulose wafers (Fig. 11.22). They are available in different shapes to fit various locations in the oral cavity. They are commonly used inside the cheeks to cover the parotid ducts. These are more absorbent than the cotton rolls or gauze pieces.

e. Suction Devices (Evacuation Systems)

Suction devices (vacuum systems) are generally of two types: High vacuum/volume evacuation system,



Fig. 11.20: Gauze piece (2″ × 2″)

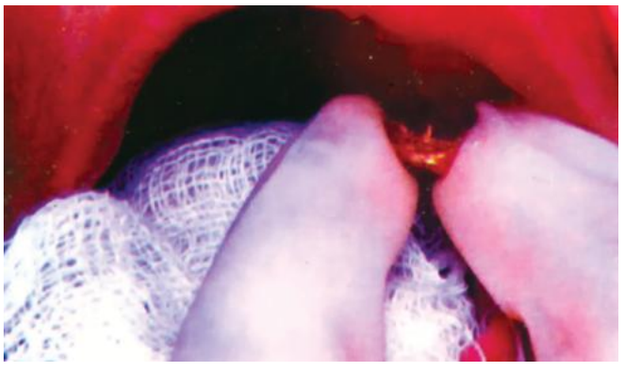


Fig. 11.21: Throat screen used to prevent accidental ingestion/aspiration of indirect restoration



Fig. 11.22: Cellulose wafers/cheek pads

which is generally operated by the dental assistant; whereas low vacuum/volume evacuation system is attached to the saliva ejector and may remain in the oral cavity during the operative procedure.

i. *High Volume/Vacuum Evacuators*

When using a high speed handpiece, both air and water emerge from the head of the handpiece to wash the working area and to act as a coolant for the bur and the tooth. High volume evacuators are preferred to remove this collected moisture and debris in the oral cavity (Fig. 11.23). High volume evacuator has the ability to clear 150 mL of water in approximately one second.

The high volume evacuator tips are usually made of disposable plastic or autoclavable metals. The tip is usually beveled and is placed intermittently in the oral cavity during the operative procedure. The evacuator tip is placed as near as possible to the tooth being prepared, but should not interfere with the operator's access or vision. It should not be positioned close to the handpiece head; water supply may get diverted away from the rotary instrument. The tip of the evacuator should be placed distal to the tooth being prepared. For ease of manipulation, the assistant holds the evacuator tip in his right hand and the air water syringe in his left hand.

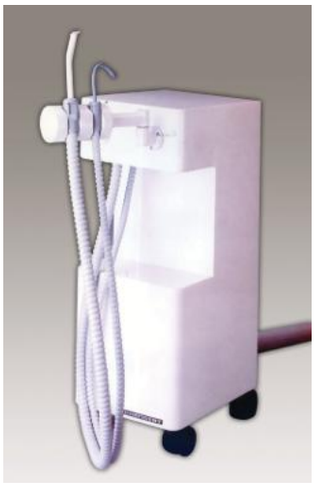


Fig. 11.23: High volume evacuator and saliva ejector

Advantages

- Removes shavings of tooth and restorative material as well as other debris from the working site.
- Toxic material is readily removed.
- Decreases treatment time as intermittent rinsing and washing is avoided.

ii. *Low Volume/Vacuum Evacuators*

Low volume evacuators are basically saliva ejectors, which are meant to remove the saliva that collects on the floor of the oral cavity. They are available with disposable plastic tips (Fig. 11.24) or autoclavable metallic tips (which should have a rubber end to prevent irritation to the soft tissue). They may be shaped by bending with fingers and are most often used along with cotton rolls, cheek pads and rubber dam.

Saliva ejectors should be placed with their tips, directed backwards and not directly in contact with the tissues. This is to prevent aspiration of the delicate mucous membrane into the holes of the tip and getting traumatized by the vacuum energy. When using saliva ejector along with rubber dam, the saliva ejector should be passed through a hole punched in the rubber dam and placed into the sulcus.

f. Gingival Retraction Cord

'*Retraction*' is the downward and outward movement of the free gingival margin, achieved by different retraction materials and the techniques.

'*Relapse*' is the tendency of the gingival cuff to revert to its original position; influenced by the elasticity of the gingival cuff and the rebound forces of adjacent attached gingiva.

'*Displacement*' is a downward movement of the gingival cuff, usually caused by heavy consistency impression materials.

'*Collapse*' is the tendency of the gingival cuff to flatten under forces associated with the use of closely adapted customized impression trays.

The cords are readymade woven cotton or synthetic fibers. Various types of cord, viz. braided/non-braided, plain/impregnated, coarse/fine are available in different sizes (Fig. 11.25A, B). The plain cords may be impregnated with chemicals before their insertion into the sulcus. Some cords are wrapped in resin wire

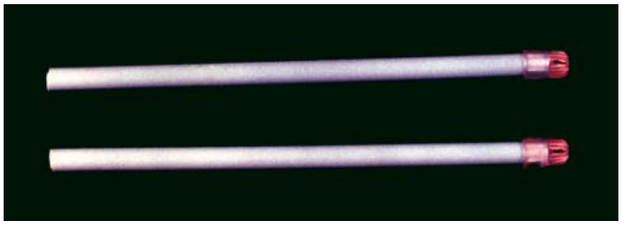


Fig. 11.24: Disposable plastic suction tips

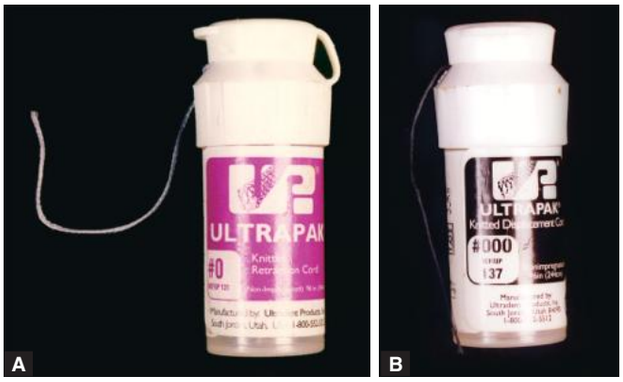


Fig. 11.25: Retraction cords: **(A)** Knitted (#0 size); **(B)** Knitted (#000 size)

to make them more compact and immobile. The cords are inserted in the gingival sulcus to keep the moisture and gingiva away from the tooth surface.

Gingival retraction cord should not be used for the displacement of gingival tissues when the tissues are swollen/inflamed. Only healthy gingiva returns to its original position after removal of the retraction cord.

Retraction cord may lead to:

- Displacement of the free gingiva laterally by few tenths of a millimetre, thus opening the sulcus.
- Apical positioning of the gingival crest, although no attempt is made to force the gingival retraction cord apically.
- Transient dehydration of the gingiva.
- Decreased bleeding (when the cord is impregnated with a vasoconstrictor, like adrenaline or a styptic like Hemodent).

Advantages

- Provides improved access and visibility.
- Protects gingiva from abrasion during cavity preparation.
- Restricts excess restorative material from pushing into the sulcus.
- Everts gingival tissue, thus exposing margins of the cavity.

The relative thickness of cords is indicated by color coding, viz.:

- Black—000 (extra thin)
- Yellow—00 (thin)
- Purple—0 (very thin)
- Blue—1 (average)
- Green—2 (thick)
- Red—3 (extra thick)

Ideal Requirements of Retraction Cord

- Should be dark in color
- Should effectively retract the gingival tissues
- Should not have any systemic effect
- Effects should be reversible
- Should not stick to the soft tissues
- Should provide effective hemostasis

Technique

In case of single cord, the cord is cut according to the need, moistened in the selected chemical agent (pre-soaked cords are also available) and inserted into the gingival sulcus area applying slight pressure at the apical direction. The instrument should be inclined towards the area already tucked by the cord.

Occasionally, it is necessary to hold the cord with one instrument while packing with the second. The instrument must be angled slightly toward the root to facilitate the subgingival placement. Excess cord is cut off in the mesial interproximal area.

In case of double-cord technique, a small diameter cord is placed in the sulcus followed by a second cord soaked in the hemostatic agent above the small

Placement of the retraction cord

- Insert cord only after anesthetizing the area.
- Choose cord that can be gently inserted into the sulcus without causing ischemia.
- The diameter of the cord should be such that it does not blanch the tissue nor is inadequate in applying pressure. If several cords have to be inserted, start with the smallest diameter one followed by the larger ones.
- Length of the cord should be such that it extends 1.0 mm beyond the gingival width of the cavity or extends around the whole circumference of the tooth.
- Avoid putting the ends interproximally. The ideal location is at the axial angles of the tooth, where the interdental col has its maximum height thus creating a better grip and stabilization on the packed cord.
- The packing instrument should be blunt, preferably with a serrated face. A cowhorn explorer or plastic instrument can also be used for the same purpose. Whatever instrument is used, the cord should be packed slowly and progressively.
- The force should be directed slightly towards the tooth surface at an angle. Apical pressure may damage the junctional epithelium.
- In shallow sulcus or when there is thin free gingiva, after inserting one end of the cord, stabilize it with a blunt instrument while the rest of the cord is packed.
- Never remove the cord dry, otherwise it may adhere to the epithelium and on pulling, cause its abrasion and profuse bleeding.
- Immediately after removal, check for pieces of gingival retraction cord that may have been torn and left in the gingival environment.
- For healthy healing of the periodontium, any object irritating the gingiva should be removed followed by oral hygiene measures.

diameter cord. Diameter of the second cord should be the largest diameter that can be placed into the sulcus. After 8–10 minutes, the second cord is soaked in water and removed. The preparation is dried and impression is made with primary cord in place. After impression making, the small diameter cord is soaked in water and removed from the sulcus.

Advantage of placing double cord is that the first cord remains in place thus, reducing the tendency of gingival cuff to recoil and displace partially set impression material. 2.0% ferric sulfate has been used to control bleeding, when directly placed at the gingival sulcus. When hemostasis is required, retraction cord soaked in the ferric sulfate solution is packed into the sulcus following conventional guidelines. It is advised to make multiple impressions in case more teeth are to be involved, especially the anteriors. Gingival retraction of alternate teeth should be carried out and impression is made; followed by retracting the gingiva of remaining teeth and impression making.

B. INDIRECT METHODS

The above mentioned measures are helpful in managing the moisture/saliva directly in the oral cavity. The measures that reduce the amount of salivation and indirectly aid in isolation are:

a. ***Comfortable and relaxed position of the patient:*** The patient should be comfortably seated in the dental chair. The surroundings should be pleasant and relaxing. These features along with comforting attitude of the dental staff reduce the anxiety levels of the patient and aid in reducing salivation.

b. ***Local anesthesia:*** Using a local anesthetic helps in reducing the discomfort and also the anxiety; subsequently controlling moisture by decreasing salivation.

The vasoconstriction caused by local anesthetic helps in reducing hemorrhage at the operating site. The salivary flow is reduced, helping in moisture control.

c. ***Drugs:*** Drugs can reduce salivation, but are rarely indicated. These include antisialogogues, anti-anxiety agents, sedatives, etc.

i. *Antisialogogues:* Premedication may be indicated using an anticholinergic agent to depress salivation. Atropine can be given half an hour before the appointment, but should be avoided in patients with high ocular pressure or with cardiovascular problems.

ii. *Antianxiety agents and sedatives:* Premedication with these drugs is quite helpful in apprehensive patients; for example, diazepam or barbiturates, 24 hours before the appointment. Because of psychological dependence on these drugs, these should be given only for short periods and to selected patients.

iii. *Muscle relaxants:* Muscle relaxants are also helpful.

II. ISOLATION FROM THE SOFT TISSUES (SOFT TISSUE MANAGEMENT)

During any operative procedure, adequate care should be taken to protect the soft tissues surrounding the tooth, which include the cheeks, lips, tongue and gingiva. Also, in order to aid in proper cavity preparation, subsequent impression procedures and restoration, soft tissue should be excluded from the operating site. Various methods of isolating soft tissues are as follows.

A. Management of Cheeks, Lips and Tongue

Various devices employed for the retraction of cheeks, lips and tongue include:

a. Rubber dam (most efficient)
b. Cotton rolls and holder
c. Tongue guards
d. Tongue depressors
e. Cheek and lip retractors
f. Mouth mirrors

a, b. ***Rubber dam and cotton rolls*** for retracting soft tissues have been discussed earlier.

c. ***Tongue guards:*** Tongue guards protect against injury to the tongue. They create a wall between the tongue and the operating field. They can be made of plastic (usually disposable) or metal (autoclavable).

d. ***Tongue depressor:*** Tongue can be manipulated and protected by using a tongue depressor, which lowers the tongue so as to avoid interference during any operative procedure. Also cheek retraction can be readily accomplished with it. Disposable wooden tongue depressors are quite popular.

e. ***Cheek and lip retractors:*** Cheek and lip retractors usually fit around the upper and lower lips including the corners of the mouth and help in pulling them backward and outward exposing the facial surfaces of maxillary and mandibular teeth. Some of these devices fit only one lip. These are used mainly for photographic purposes and when working on anterior teeth (Fig. 11.26).

f. ***Mouth mirror:*** Mouth mirror held with one hand help retracting the tongue or cheek as per need.

B. Management of Gingival Tissues

The management of gingival tissues, their retraction and deflection is mandatory for recording margin



Fig. 11.26: Cheek and lip retractor

details of the preparation in impression making. The selection of method and materials for gingival retraction mainly depend upon the clinical situations, viz. inflamed/fibrosed gingiva, hemorrhage at the site, etc. The operator should carefully select the retracting material considering potential benefits and disadvantages. Gingival retraction is defined as a process of exposing margins when making impressions of the prepared teeth. It is the deflection of the marginal gingiva away from a tooth. The gingival sulcus is deepened to expose the cervical portion of the tooth in order to have proper marginal finish to the restoration. The main goal of gingival retraction is to reversibly displace the gingival tissues, so that low viscosity impression material can be introduced into the sulcus, capturing the margin details without tearing the impression.

Need for Gingival Deflection

- To provide access for impression material into the subgingival margins, recording the finish line properly.
- Undermine caries can be treated easily.
- The margins on the prepared tooth can be evaluated and modified, if need be.
- Helps in easy removal of cement without damaging the gingiva.
- The marginal fit of the restoration can be assessed.
- To gain surface area, it may be necessary to extend the restoration below the gingival margin.
- Facilitates finishing and polishing of the restoration, especially at margins.

Indications

- Presence of subgingival caries
- Cervical abrasion and erosion
- Gingival polyp at the margins
- Subgingival finish line
- Decreased crown:root ratio

Contraindications

- Poor oral hygiene
- Gingival recession
- Bone loss
- Systemic diseases affecting gingiva

The features to be considered prior to gingival retraction are:

- The periodontium should be sound, especially around the prepared tooth and the adjoining teeth.
- The crest of free gingiva should be at its normal position relative to the tooth surface with no recession.
- Any hyperplastic/fibrosed tissue, if present, should be treated prior to restoration.
- Crevicular fluid and bleeding should be arrested to achieve accessibility and effective manipulation.
- The retraction at the gingival crevice area should be accomplished without detaching the apically located epithelial attachment and periodontal ligament.
- Gingival retraction should not cause any irreversible damage to the gingiva/periodontium.

Methods

The methods commonly used for gingival retraction are:

a. Physico-mechanical method
b. Chemo-mechanical method
c. Electrosurgical means
d. Surgical means
e. Miscellaneous agents

a. *Physico-mechanical methods:* The gingiva is displaced mechanically both laterally and apically away from tooth surface. The mechanical methods are generally employed where the gingiva is healthy with good vascular supply and adequate bone support. The various materials used for mechanical retraction are:

i. Wooden wedges (used for depressing the interproximal gingiva)
ii. Cotton twills (forcibly introduced into the gingival sulcus; mechanical retraction achieved in 20–30 minutes)
iii. Cotton twills with fast setting zinc oxide eugenol (Thin mix of zinc oxide eugenol wrapped in cotton rolls is inserted into the gingival sulcus; provides better retraction and effective tissue tolerance, however, if kept for longer time, leads to loss of periodontal attachment.)
iv. Rubber dam material (heavy—0.010 inch or 0.25 mm, extra heavy—0.012 inch or 0.30 mm and special heavy—0.014 inch or 0.35 mm) is usually employed: Effective in limited number of teeth; however, cannot be used with elastomeric impression materials.

v. Copper band (Slight oversized band filled with modeling wax is effective in pushing the gingival sulcus; excess pressure should be avoided.)

b. *Chemo-mechanical methods:* Certain chemical agents soaked in cotton or cords are utilized to achieve effective gingival retraction (retraction cord is explained in previous pages). The commonly used chemical agents are:

i. Epinephrine/norepinephrine (vasoconstrictors)
ii. Alum/aluminium chloride/tannic acid (biological fluid coagulants)
iii. Zinc chloride/silver nitrate potassium hydroxide (surface tissue coagulants)
iv. Concentrated zinc chloride/concentrated potassium hydroxide (chemical cautery)

i. *Epinephrine/norepinephrine (vasoconstrictors):* The chemical agents physiologically restrict the blood supply by decreasing the size of the blood capillaries (vasoconstriction), producing hemostasis. The retraction is achieved in 10 minutes; however, their use should be limited because of their systemic effects. It is established that the amount of epinephrine absorbed from 2.5 cm of retraction cord during 5–15 minutes in gingival sulcus is 71 µg.

Contraindications:

- Hypersensitive to epinephrine
- Cardiovascular disorders
- Hypertension
- Hyperthyroidism
- Diabetes
- Patients on β-blockers and antidepressants

ii. *Alum/aluminum chloride/tannic acid (biological fluid coagulants):* Biologic fluid coagulants locally coagulate blood and tissue fluids, effectively retracting the surface layer. 10% alum solution is considered superior to epinephrine. 20–40% tannic acid is effective and shows good tissue recovery.

iii. *Zinc chloride/silver nitrate/potassium hydroxide (surface tissue coagulants):* Usually 6–8% zinc chloride is used to coagulate the surface layer of sulcular and free gingival epithelium along with any fluids present in sulcus; however, these agents may cause ulceration and necrosis, when applied for prolonged time.

iv. *Concentrated zinc chloride/concentrated potassium hydroxide (chemical cautery):* 40% zinc chloride is used to chemically cauterize the local gingival tissues.

c. *Electrosurgical means:* Electrosurgical means implies surgical reduction of sulcular epithelium by using electrode of high frequency. Electrosurgery produces a controlled tissue destruction to achieve the desired results.

Various types of electrosurgical electrode are used in routine; such as, coagulating, diamond loop, round loop, straight loop, small loop, etc. The electrosurgical actions can be:

i. Electrosectioning/cutting (precise cutting with minimum tissue involvement)
ii. Electrocoagulation (create coagulation of tissues, their fluids and oozed blood)
iii. Fulgeration (involve deeper tissues, accompanied by carbonization)
iv. Desiccation (massive tissue involvement; uncontrolled action)

Electrosurgery: Guidelines

- The operational area should be moist.
- Use fully rectified, undamped current with minimum energy output.
- Use light pressure touch and rapid deft strokes with a five seconds lapse between two strokes (always keep cutting electrode in the internal wall of sulcus, maintaining biologic width).
- Metallic restorations should not be touched (can create short circuit and damage surrounding structures).
- Always clean debris on the electrode tip with alcohol-soaked gauge.
- After the impression/restoration procedure creates a blood clot with curetting the area.
- Avoid electrosurgery in patients with pacemaker.

Advantages

- Immediate hemostasis
- Better healing of the tissues
- Improved accuracy of the impression (margins visible)
- Decreased chair time and stress
- Minimal postoperative discomfort

Disadvantages

- Technique sensitive
- May lead to apical migration of marginal gingiva.
- Inaccurate current setting may cause tissue damage.
- Difficult to control lateral dissipation of heat

d. *Surgical means*

i. *Surgical curretage:* A surgical curettage using rotary instruments is carried out to remove epithelial tissue in the sulcus along with creating a chamfer finish line in the preparation. The technique is also known as '*gingitage*' and is used where adequate gingiva is present and the sulcular depth is less than 3.0 mm.

Disadvantage

- There is poor tactile sensation when using diamond points on gingival sulcular walls.
- May lead to destruction of periodontium.

ii. ***Gingivectomy:*** Gingivectomy, the excision of the gingiva, has been preferred to remove hypertrophic gingiva.

Indications

- Interfering gingival tissue during any impression/restorative procedures.
- Gingival polyps seen in proximal caries.
- For crown lengthening procedures.
- For apical repositioning of whole periodontal attaching apparatus to create a healthy, safely manipulated, easily retractable free gingiva.

e. ***Miscellaneous agents:*** The agents used to achieve effective gingival retraction are:

1. Magic foam
2. Merocel strips
3. Expasyl
4. Gingitrac (centrix)
5. Lasers
6. Stay-put retraction cord
7. Comprecap
8. Racegel
9. Matrix impression system
10. Cryosurgery
11. Newer agents

1. ***Magic foam:*** Magic foam is the expanding Vinyl polysiloxane material, designed for easy and fast retraction of the gingival sulcus without the potentially traumatic and time consuming packing of retraction cord. It is a non-hemostatic cordless retraction system consisting of foam and cartridges. The material in a syringe is delivered around the crown margins and a cap is placed to maintain pressure. After five minutes, the cap and the foam are removed and the preparation is ready for final impression.

   Advantages

   - Non-traumatic; application with minimum pressure
   - Easy and fast application
   - Comfortable to the patient
   - Efficient, especially in multiple preparations

   Disadvantages

   - No hemostatic action
   - Expensive

2. ***Merocel strips:*** Merocel strips (hydroxylate polyvinyl acetate) are synthetic material that expands by absorption of oral fluids and exerts pressure on surrounding tissue. Placement of strips does not require local anesthesia; ensure effective gingival retraction. The absence of fibers in the strips decreases the risk of postoperative complications. Merocel is soft and adaptable to surrounding tissues, not abrasive and can easily be shaped according to need.

   Advantages

   - Can be easily shaped and adapted around tooth.
   - Effective in absorption of oral fluids.
   - Significantly superior to retraction cords
   - Strips are free of any fiber or debris (not damaging to tissues)

3. ***Expasyl:*** Expasyl is a chemo-mechanical technique for gingival retraction. The Expasyl material is kaoline incorporated into an organic binder with aluminum chloride as hemostatic agent. It is supplied in syringe as viscous paste, which is injected into gingival sulcus, exerting a stable and non-damaging pressure. When left in place for one minute, this pressure is sufficient to achieve 0.5 mm of gingival retraction.

   Advantages

   - Effectively achieve hemostasis
   - Minimum pressure required (atraumatic)
   - Minimal tissue damage
   - Helps in controlling bleeding
   - Sufficient retraction in two minutes
   - Less time consuming

   Disadvantage

   - Thickness of paste pose problems in pushing into sulcus
   - Not cost-effective
   - Metal tips may not be suitable for interproximal areas

4. ***Gingitrac (centrix):*** Gingitrac is available as paste system; applying the paste around the margins using preloaded syringe. The paste contains aluminum sulfate, a mild natural astringent. It utilizes patient's bite pressure to push material into gingival sulcus and achieve retraction. A cap can be used in single tooth preparation. The paste is removed prior to impression making.

   Advantages

   - Works in less than five minutes
   - Gently retracts the gingiva with no tissue trauma
   - Natural astringent controls bleeding
   - Works on single or multiple crown preparations

5. ***Lasers:*** Soft tissue lasers have been tried for gingival retraction using flexible optical fibers. Lasers work through photoablation and produce blood-free incisions followed by rapid healing without any inflammation. The laser technique is a little slower; however, produces controlled tissue retraction, free of hemorrhage and pain.

The lasers commonly employed are:

- *CO_2 (carbon dioxide):* Absorb loss energy near the tissues, with minimal increase of temperature.
- *Nd: YAG (Neodymium:yttrium-aluminum garnet):* Preferred retraction of hypertrophied tissues
- *Er:YAG (Ebrium:yttrium-aluminum garnet):* Fairly safe (not good hemostasis)
- *Argon:* Safe to use.

Advantages

- Minimal or no anesthesia; relatively painless.
- Do not harm dental tissues, including pulp.
- Can be used around dental implants.
- Carbon dioxide laser provides better hemostasis.
- Healing is rapid.

Disadvantages

- Er:YAG is not good for hemostasis.
- May not provide tactile feedback (CO_2 laser), leading to damage to attached gingiva.

6. ***Stay-put retraction cord:*** Stay-put is a unique combination of softly braided retraction cord and ultra-fine copper filaments. When the stay-put cord is shaped, it remains in shape and does not deform. It is usually available as son-impregnated; however, aluminum chloride can be used for impregnation.

   Advantages

   - Can be easily adapted.
   - Can be shaped.
   - Does not deform in the sulcus.
   - Non-impregnated; can be impregnated with an astringent or hemostatic solution, if need be.

7. ***Comprecap:*** The use of aluminum chloride in conjunction with comprecap compression caps, is effective in controlling bleeding and providing tissue retraction. It holds the retraction cord deep in the sulcus, opening it even wider. After placing the retraction cord, the cap is placed over the prepared tooth and pushed into the sulcus. Patient is asked to bite on the cap for sometime; after that it is removed along the retraction cord.

   Advantages

   - Ensures a dry and clean area.
   - Less technical skill required.
   - Stops bleeding naturally by compression.

   Disadvantage

   - Aluminum chloride used in conjunction with comprecap may interfere with setting of polyether/polyvinyl siloxane impression material.

8. ***Racegel:*** Racegel, a new hemostatic agent, available in gel form, controls bleeding and absorbs crevicular fluid prior to impression making. The gel contains 25% aluminium chloride, oxyquinol and excipients. Racegel can be used with or without gingival retraction cords. The material's viscosity increases when in contact with the gingival tissue providing access to the gingival margin (thermodynamic property).
   It facilitates access of the gingival crevice, reduces bleeding and oozing. Due to its consistency, it can be rinsed quickly, leaving no residual material.

   Advantages

   - Effective hemostatic agent.
   - Can easily be removed.
   - Can be used with or without retraction cord.

   Disadvantages

   - May irritate eyes and skin.
   - Allergic reaction reported because of oxyquinol.

9. ***Matrix impression system:*** The matrix impression system requires series of three impressions using three viscosities. A matrix of occlusal registration rubber base material (semi-rigid) is made over the tooth preparation prior to gingival retraction.
   The matrix is trimmed as per need and the retraction cord is removed. An impression is made with high viscosity rubber base impression material. After seating the matrix impression, a stock tray filled with medium viscosity impression material is seated over the matrix and the remaining teeth to make impression of the entire arch. Matrix impression system maintains retraction by trapping a highly viscous material in the sulcus when the matrix is seated.

   Advantages

   - Eliminates chances of tearing of sulcular flange.
   - Cleans blood and debris from sulcus area.
   - Delivers impression material in sulcus with accuracy.
   - Open sulcus for longer period of time.

   Disadvantage

   - Increased chairside time.

10. ***Cryosurgery:*** Cyrosurgery is preferred in case of interfering and fibrosed gingival tissues. It is also indicated for apical repositioning of whole periodontal apparatus to create a healthy, retracted free gingiva. A sharp, cold knife is used to remove the tissues at 320°–340°F using liquid nitrogen. The tissues freeze and shrink when in contact with nitrogen.

    Advantage: Effective in fibrosed gingival tissues.

11. ***Newer agents:*** Nasal and ophthalmic decongestants show favorable results in gingival retraction. The commonly used agents are:

    - 0.25% phenylephrine hydrochloride (neosynephrine)

- 0.05% oxymetazoline hydrochloride (afrin)
- 0.05% tetrahydrozoline hydrochloride (visine)

Their use is limited because other techniques are more effective in achieving gingival retraction.

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